

Review - Agriculture, Agribusiness and Biotechnology

Unlocking Sustainable Agriculture: Direct and Indirect Effects of *Pseudomonas* Genus on Crop Yield Enhancement

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HIGHLIGHTS

- *Pseudomonas* assessed as a genus with great potential for agriculture use.
- Metabolites from pathogenic *Pseudomonas* are an alternative for agricultural inputs.
- Agricultural bioinputs containing *Pseudomonas fluorescens* are marketed in Brazil.

Abstract: *Pseudomonas* is a large bacterial genus consisting of more than 300 species commonly found in the environment and isolated from soil, water, and plants. This species has a great affinity for grasses and can colonize the interior of their roots, which characterizes them as endophytic rhizobacteria. Endophytic microorganisms are beneficially associated with the host and exchange nutrients necessary for mutual growth and development. Endophytic *Pseudomonas* promote plant growth directly via the process of biological nitrogen fixation, solubilization of phosphate and other macro and micronutrients, and synthesis of siderophore and phytohormones, and indirectly by protecting against pathogens through the induction of systemic resistance or release of volatile and non-volatile metabolic compounds. Representatives of the genus *Pseudomonas* have been extensively studied for their associative characteristics. Products formulated with *Pseudomonas* are commercially available, of which 12 are registered in Brazil. However, some *Pseudomonas* species, although promising for agriculture, can be pathogenic, for example, *P. putida* and *P. fluorescens* can cause diseases in debilitated humans and *P. syringae* is pathogenic to plants. In this scenario, alternative molecular and biotechnological studies on the exclusive use of metabolites might represent an alternative that deserves efforts for improvement. This article reviews the potential of *Pseudomonas* genus members to sustainably enhance agricultural yields and also highlights alternatives for the use of pathogenic species.

Keywords: Bioinputs; PGPB; *Pseudomonas*; Sustainable Agriculture.

INTRODUCTION

Plants, the primary producers in the world's ecosystems, often owe much of their growth success to unseen allies, the microorganisms. Some plant growth-promoting microbes form a mutually beneficial relationship with their host plants [1]. Bacteria constitute the most diverse and significant groups of microorganisms on Earth, with plant growth-promoting bacteria (PGPBs) occupying key roles in the rhizosphere, phyllosphere, and endosphere. Their benefits make them viable, environmentally friendly alternatives to chemical fertilizers throughout plant development [2].

Bacteria that are compatible with the plant establish a beneficial relationship, marked by processes such as signal recognition or neutralization of potential disease-causing entities. These interactions ultimately contribute to the overall health and development of the plant. Given these capabilities, PGPBs are increasingly being recognized as sustainable alternatives to chemical pesticides, offering a promising route to environmentally friendly and effective plant protection strategies [3].

The genus *Pseudomonas* of the Proteobacteria group encompasses more than 300 species and is a group of Gram-negative, rod-shaped bacteria isolated predominantly from plants, soil and aquatic environments. *Pseudomonas* is known for its rapid growth rate and extensive metabolic versatility, making it particularly advantageous for various biotechnological applications [4,5]. Diverse species of the genus have been identified as bioremediation agents and hosts for the industrial bio-manufacturing of natural products, such as peptides, terpenoids, biopolymers, and enzymes [6,7].

This genus is frequently found in relations with plants, including maize (*Zea mays* L.), wheat (*Triticum* spp.), sugarcane (*Saccharum officinarum*), and rice (*Oryza sativa*). Certain associations have been observed with legumes such as peanuts (*Arachis hypogaea*), soybeans (*Glycine max* L.), and beans (*Phaseolus vulgaris* L.), albeit less significantly than those with grasses [1]. Impressively, *Pseudomonas* genus represented 64.2% of the bacteria isolated from the rhizosphere and roots of four representative grass species cultivated in China [8], given that demonstrates the potential of this group to become a new possibility for agricultural bioproducts.

Pseudomonas spp. isolated from the roots of different grass species may exhibit numerous growth effects, via solubilizing organic and inorganic phosphate, synthesizing indole acetic acid (IAA), and regulating ethylene production through the enzyme 1-aminocyclopropane-carboxylic acid deaminase (ACC deaminase). They may also protect plants against phytopathogenic fungi using volatile compounds and contribute to phytoremediation by degrading petroleum hydrocarbons [9,10].

Several *Pseudomonas* species often considered opportunistic human pathogens [11,12], have shown promising activity in plant-growth promotion and in the biocontrol of phytopathogens. For example, *P. putida*, a producer of volatile organic compounds [13] and *P. fluorescens*, which may promote plant growth by increasing plant tolerance to water stress, or by solubilizing phosphate and synthesizing siderophores [14], may cause skin and soft tissue infections in humans. As a result, there are regulatory restrictions on the production and sale of products derived from these species, with rigorous environmental and human health assessments required before product approval [15]. *Pseudomonas* species characterized as human pathogens are listed in Table 1.

This paper is an integrative review article that seeks to address the main mechanisms of action of the genus *Pseudomonas* in aiding plant protection and growth, with a focus on agricultural sustainability, enabling the reader to better understand the importance of these bacteria for increasing the productivity of plant-based foods, in addition to offering examples of commercialized products and perspectives for the exploration of strains not yet used.

Table 1. List of *Pseudomonas* species considered pathogenic to humans

Species	Classification on Year	Disease	Validly name by LPSN 2024 [5]	Reference
<i>P. aeruginosa</i>	1872	Nosocomial infections	Yes	[16]
<i>P. alcaligenes</i>	1928	Bloodstream infection	Yes	[16,17]
<i>P. andersonii</i>	2001	Pulmonary granuloma	No	[18]
<i>P. asiatica</i>	2019	Hospital clinical samples	Yes	[19]
<i>P. fluorescens</i>	1895	Skin and soft infection	Yes	[16,20]
<i>P. fulva</i>	1963	Bloodstream infection	Yes	[21]

Cont. Table1

<i>P. japonica</i>	2008	Skin and soft infection	yes	[22]
<i>P. juntendi</i>	2019	Hospital clinical samples	yes	[23]
<i>P. luteola</i>	1985	Endocarditis	<i>Chryseomonas luteola</i>	[24]
<i>P. mendocina</i>	1970	Bacteremia	yes	[25]
<i>P. montellii</i>	1997	Hospital clinical samples	yes	[26]
<i>P. mosselii</i>	2002	Clinical samples	yes	[27]
<i>P. nosocomialis</i>	2019	Hospital clinical samples	<i>Stutzerimonas nosocomialis</i>	[28]
<i>P. oryzihabitans</i>	1985	Postoperative septicemia	yes	[29]
<i>P. otitidis</i>	2006	Otic infections	yes	[30]
<i>P. pickettii</i>	1973	Pseudobacteremia	<i>Ralstonia pickettii</i>	[31]
<i>P. poae</i>	2003	Septic transfusion	yes	[32]
<i>P. pseudoalcaligenes</i>	1966	Peritonitis	<i>Pseudomonas oleovorans</i>	[16,33]
<i>P. putida</i>	1889	Skin and soft infection	yes	[16,22]
<i>P. stutzeri</i>	1896	Endocarditis, endophthalmitis	yes	[16,34]
<i>P. yangonensis</i>	2020	Hospital clinical samples	yes	[35]

¹ This table was generated from the summary of clinical cases reviewed by Bartlett and coauthors [36].

The genus *Pseudomonas*: Overview and Classification

First described in 1894 through morphological analysis by Migula, the genus *Pseudomonas* was validated in the Approved Lists of bacterial names in 1980 [37]. Species of this genus are ubiquitously distributed, being isolated from various organisms and environments including animals, plants, algae, fungi, air, soil, water, glaciers, and deserts [38].

Evolution of *Pseudomonas* Taxonomy and Its Implication

The evolving use of phylogenomic technologies has led to significant changes in bacterial taxonomy, including dividing genera into subgenera or creating new genera, and has notably increased the number of identified species, such as in the case of *Pseudomonas* [39]. This can be observed in the increasing number of *Pseudomonas* species identified in the past few years—from approximately 250 species in 2021 [4] to more than 300 validated in 2024 by the List of Prokaryotic Names with Standing in Nomenclature [5]. Of the newly described species, 43 were identified by sequencing the complete genomes of the investigated strains [40].

The high diversity within the *Pseudomonas* genus contributes to its adaptability and success in the colonization process of host plants, in different environmental conditions. *Pseudomonas* genera is frequently isolated from host plant species, and both are usually reported for their beneficial effects on the plant hosts, whether direct or indirect [41-43].

Pseudomonas and Plant Growth: Direct Mechanisms

Direct mechanisms include any resource that the bacterium has to directly assist in plant growth, either through the generation of plant growth regulators (growth phytohormones) or by biofertilization, with increased nutrient uptake, especially of nitrogen and phosphorus [44,45]. Nutrient Uptake: The ability of the *Pseudomonas* genus to make nutrients and metabolites associated with plant growth promotion available

The nutrient uptake from the action of *Pseudomonas* encompasses various mechanisms. Firstly, it involves the fixation of atmospheric nitrogen secondly, it supports the acquisition of nutrients from the free-living phase in the soil, which are subsequently extracted and transferred to the plants through oxidative processes. Third, it facilitates the cycling of nutrients from decomposed plant material and insects, thereby enhancing their availability and improving soil fertility [42]. And finally, the increase in the absorption area by the roots through the action of phytohormones, promoting greater use of these nutrients [46]. This complex interaction between *Pseudomonas* and plants has a profound influence on plant growth and productivity.

Pseudomonas, although not commonly known for its nitrogen-fixing abilities, has been identified as a free-living bacterium proficient in this aspect. These microorganisms inhabit the rhizosphere, derive energy from organic matter decomposition, and can transfer nitrogen without intimate plant contact [47]. *Pseudomonas stutzeri* A1501, which was originally isolated from the rhizosphere of rice, is an example of *Pseudomonas* sp. that can colonize the root surface, invade the radicular tissues of host plants

endophytically, and fix nitrogen by forming biofilms. In the biofilm state, bacteria show many advantages compared to those in the planktonic state, such as increased resistance, protection from environmental stresses, and adaptation to nutrient deprivation [48,49].

When this strain is subjected to a mutation in the *nifH* gene (*P. stutzeri* A1502 *nifH*⁻) and inoculated into maize, colonized the roots and increased biomass gain by 25.4% and N concentration by 7.8%. The nitrogen fixation of the mutant strain used in this study was confirmed by the ¹⁵N dilution technique, demonstrating its importance for plant growth and its ability to reduce synthetic fertilizer usage. However, these results were obtained in a hydroponic system, whereas in the soil, the contribution to growth was lower, which was attributed to soil characteristics and the existing microbial community [48].

Engineering studies have further expanded the utility of *Pseudomonas* for nitrogen fixation. For instance, *P. protegens* Pf-5, known for its biocontrol properties, was engineered with the nitrogenase (*nif*) island from *P. stutzeri* DSM4166 to create a recombinant strain exhibiting both biocontrol properties and nitrogen fixation ability. This recombinant strain showed significant nitrogenase activity, comparable to the wild-type DSM4166 strain [50]. Similarly, another engineered strain of *P. protegens*, CHA0, demonstrated promising plant growth promotion when introduced into *Arabidopsis thaliana* [51].

Moreover, *P. fluorescens* has been shown to promote maize growth, resulting in a 29.8% increase in grain yield with a 25% reduction in nitrogen fertilizer usage in a field experiment. The strain CCTB03 used in this study is a part of the formulation of a new commercial inoculant developed in Brazil [52]. Even if the strain used is not a nitrogen-fixing bacteria, reducing the use of nitrogen fertilizer contributes to soil conservation, preventing the accumulation of this nutrient. These studies demonstrate that genetic engineering can be a valuable tool in utilizing *Pseudomonas* strains to supply nitrogen to crops, either by enhancing a beneficial species such as *P. protegens* or by enhancing and silencing pathogenicity genes, as seen in the example of *P. stutzeri*.

Another essential nutrient for plant development is phosphate, although is the second most abundant nutrient in the soil, only a small percentage is bioavailable to plants because of binding to soil metals, such as Al, Fe, and Ca, forming aluminum phosphate, iron phosphate, and calcium phosphate, respectively [53]. Phosphate solubilizing bacteria (PSB) can mitigate this limitation by increasing plant uptake of phosphorus. These bacteria have the potential to mineralize and solubilize low-solubility organic, and inorganic phosphorus, and convert them into plant-available phosphate species, such as orthophosphate [53,54].

Different species of *Pseudomonas* (*P. proteolytica*, *P. palleroniana*, and *P. azotoformans*) were reported as being able to release organic acids such as oxalic, malic, lactic, citric, and succinic acids in analysis performed by HPLC. The presence of these acids contributes to the growth of *A. thaliana* [55]. *Pseudomonas jessenii* proved to be a PSB by solubilizing 398.14 µg mL⁻¹ of phosphate when inoculated in liquid medium, in addition to promoting the germination of 92% of chickpea seeds (*Cicer arietinum*) *in vitro*, corresponding to an increase of 22% when compared to the control [56].

PSBs also utilize enzymes, such as phosphatase, to mineralize phosphate [54]. Organic compounds of animal or plant origin that are present in the contains a variety of phosphoric components, such as phospholipids, nucleic acids, and polyphosphates which are hydrolyzed by PSBs through the action of alkaline and acid phosphatases, such as phytases, releasing inorganic phosphorus that is immobilized by the plant [57].

Pseudomonas aeruginosa ATC 15442 known to exhibit phosphatase and phytase enzyme activities, was utilized in a study by Ortega-Torres and coauthors [58] to convert organic phosphorus into inorganic phosphorus in bovine compost. The successful release of 94.8% plant-assimilable nutrients inspired suggestions to produce this enzymatic mixture on an industrial scale for agricultural use.

The use of a transgenic strain of the root bacterium *P. simiae* with gene insertion for the enzyme phytase showed increased bacterial efficiency in hydrolyzing phytate through the action of organic acids capable of releasing orthophosphate. Remarkably, upon introducing this strain into *A. thaliana*, significant plant growth was observed when phytate was the only available phosphate source, highlighting the potential of synthetic biology in overcoming phosphate solubilization challenges [59].

The ability of *Pseudomonas* to increase phosphate solubilization in soil serves as a powerful tool for promoting plant growth and increasing crop productivity. The mechanisms behind this increase involve the production of organic acids and enzymes, which facilitate the mineralization, solubilization, and immobilization of phosphate. With the emergence of new technologies, such as synthetic biology, the exploitation of these mechanisms can be improved. Among the various species mentioned, *P. aeruginosa* stands out, which, although pathogenic, can have great value for the bioinput market if used in an inactive form in a product containing only the enzymes phosphatase and phytase.

In addition to the nutrients mentioned, phytohormones, as crucial biochemical intermediaries, exert profound influence on plant metabolism. Their roles range from growth regulation to defense activation against stressors, and they can act as signaling triggers for the synthesis of secondary metabolites to protect plants against environmental stresses such as drought and excess salinity [60-62]. Plant growth is primarily regulated by five classes of phytohormones: auxins, gibberellins, cytokinins, ethylene, and abscisic acid. The microbiota that inhabit the rhizosphere can synthesize primary hormones and make them available to the plant, thereby affecting plant development by inducing or suppressing growth [46].

Auxin influences numerous aspects of plant development, including the formation of vascular tissue, stomata, and trichomes, which are critical in leaf development [63], its modulate seed production, are involved in the progression of the endosperm and tegument, and participate in the induction, regulation, and maintenance of primary dormancy and root development [64,65].

Indole-3-acetic acid (IAA), the most common auxin in plants. IAA is synthesized from tryptophan primarily through the indole-3-pyruvic acid (IPyA) pathway, a main biosynthetic route in terrestrial plants [64,66].

A prime example of a bacterial species capable of synthesizing IAA is *P. moraviensis*. When inoculated in wheat, this bacterium led to a remarkable 33% increase in IAA in the rhizosphere soil compared to the uninoculated control [67]. When tryptophan was added to the bacterial inoculum, the IAA content in the soil increased to 52%. The authors also mutated the bacterium to generate a strain with a deficiency in IAA conversion. When the bacterium was genetically modified to impair IAA conversion, it ceased using tryptophan, halting the release of the hormone into the plant and thus stunting its growth, illustrating the vital role of the tryptophan pathway in bacterial metabolism.

In vitro studies have shown that *P. fluorescens*, isolated from non-rhizospheric soil, demonstrated maximum IAA production in a medium containing L-tryptophan as a hormone precursor and fructose as a carbon source [68]. The inoculation of *P. fluorescens* and *P. putida* in *A. thaliana* increased lateral root formation, root hairs, and plant biomass due to auxin induction [69]. Furthermore, the production of IAA by *P. stutzeri* promoted the growth of onions (*Allium cepa* L.) [70].

Conversely, auxin can also be used as a virulence factor by some phytopathogenic strains, as exemplified by *P. syringae* DC3000, which uses auxin to suppress the defenses of *A. thaliana*, thereby promoting pathogenicity [71]. Further investigation has shown that IAA also regulates the genes of this bacterial strain, affecting not only virulence factors but also the expression of stress-response genes, thereby assisting bacterial survival under unfavorable conditions [72].

To summarize, auxins, particularly indole-3-acetic acid (IAA), play a multifaceted role in plant growth and development. Species little explored in agricultural formulations, such as *P. moraviensis* and the pathogenic *P. putida* and *P. stutzeri*, become interesting for new studies in the area of large-scale biotechnological production of plant hormones.

In iron deficiency, the plant experiences serious damage to its growth, leading to symptoms such as chlorosis due to interrupted photosynthesis [73]. Siderophores, low-molecular-weight organic compounds produced by microorganisms and some plants, may have an important role under conditions of iron deficiency. Siderophores of bacterial origin have been utilized in agriculture to promote plant growth and bioremediation in the form of biofertilizers, biofungicides, and as biosensors detecting environmental iron levels [74,75]. Rhizobacteria capable of producing these compounds facilitate plant growth by providing soluble iron to the host and allowing protection from pathogenic fungi by decreasing the availability of iron in the root region [76].

Pseudomonas putida isolated from mung bean rhizospheric soil showed 85.3% activity in the chelation of iron via siderophores in iron-deficient soils, allowing its use as biofertilizer for this legume species [89]. Still, no product sold in Brazil that uses *Pseudomonas* in bean crops [77].

Pseudomonas japonica is a siderophore-producing bioremediation agent that promotes the survival of alfalfa (*Medicago sativa*) when inoculated with aluminum, lead, or cadmium *in vitro*. In addition to their affinity for iron, siderophores also have an affinity for heavy metals and form stable complexes with them; therefore, they are of great use in biotechnological processes involving bioremediation [78]. Already in maize (*Zea mays*), *Pseudomonas* sp. and *P. fluorescens* increased the vegetative growth parameters of plants grown in soil contaminated with copper by providing siderophores [79].

In summary, siderophore-producing *Pseudomonas* species are vital components in the plant microbiome, facilitating the availability of iron and potentially aiding in the sequestration of heavy metals. Their application in agriculture could have significant implications for enhancing crop growth and resilience in challenging environmental conditions.

Although the macronutrients and phytohormones mentioned are extremely important to crops, micronutrients, while required in smaller quantities, are essential for optimal plant growth and metabolism.

The deficiency of micronutrients such as zinc (Zn), boron (B), nickel (Ni), manganese (Mn), molybdenum (Mo), and chloride (Cl) can lead to a host of issues in plants, including disrupted enzymatic function, cell damage, oxidative stress, metabolic disturbances, and loss of productivity [80].

Zinc is the most studied micronutrient and it is highly deficient in most soils. Interestingly, some genera of bacteria, such as *Acinetobacter*, *Bacillus*, and *Pseudomonas* can solubilize zinc. A Zn-solubilizing

P. fluorescens strain was isolated by Di Simine and coauthors [81] from forest soil. A strain of *P. aeruginosa* was isolated from the rhizosphere of sugarcane agricultural fields by Devi and coauthors [82] also demonstrated a zinc-solubilizing capacity *in vitro*. Vaid and coauthors [83] isolated another *Pseudomonas* strain with the ability to solubilize Zn, increasing the number of total panicles and grain weight when inoculated into rice.

Zaheer and coauthors [84] demonstrated the high Zn-solubilization capacity of the AZ5 strain of *Pseudomonas* sp. The mechanism for improving Zn availability is chelation, Zn-chelating compounds increase the bioavailability of zinc in the rhizosphere; this mechanism found in *P. montellii* [85]. Inoculation of Zn-solubilizing bacteria in zinc-deficient soils decreased the symptoms of Zn deficiency in the plant and increased the total biomass, grain yield, and root and shoot weights in rice [86]. Inoculation of maize seeds with the P29 strain of *Pseudomonas* sp. enhanced Zn and other macro- and micronutrient concentrations [87].

Manganese is another micronutrient that is made bioavailable through the action of endophytic microorganisms. Genera such as *Acidovorax*, *Comamonas*, *Rhizobium*, and *Pseudomonas* exhibit Mn-oxidizing activities and can form Mn biofilms and biogenic Mn oxides. However, studies on the mechanisms underlying these effects are rare. Zhao and coauthors [88] reported that *Pseudomonas composti* strain SS02, isolated from the wetland plant *Suaeda salsa*, has Mn-oxidizing capacity, and the strain promoted plant growth.

In short, the role of *Pseudomonas* species in solubilizing and oxidizing essential micronutrients is well documented. By harnessing their unique capabilities, we can potentially alleviate micronutrient deficiencies and bolster plant growth, enhancing agricultural productivity and sustainability.

Given the above, when analyzing commercial products registered for promoting plant growth in Brazil, one can note the presence of *P. fluorescens* in isolated formulations or co-cultivation with other microorganisms, such as *A. brasilense*, *Bacillus amyloliquefaciens*, and *Priestia megaterium*. These products are produced by several companies for use in rice, corn, forage grass, and soybean crops [77]. Even so, one can observe the opportunity in the development of new products involving the different species of *Pseudomonas* mentioned in this section to promote the growth and productivity of crops not yet covered, such as alfalfa, chickpea, wheat, triticale, cucumber, and onion (Table 2) in addition to all the diversity not mentioned here.

***Pseudomonas* and Plant Growth: Indirect Mechanisms**

Indirect mechanisms employed by PGPB such as *Pseudomonas* provide protection for plants against a spectrum of biotic and abiotic stresses [44]. They primarily function as biocontrol agents against phytopathogens employs two main strategies. First, they exert local antagonistic actions, such as releasing metabolites that inhibit pathogen growth or rendering their survival untenable. Second, they induce systemic resistance in plants, thereby increasing the plant's inherent defense capabilities [45]. When confronted with pathogenic threats or abiotic stressors, plants utilize their intrinsic defense mechanisms bolstered by bacteria-mediated defense responses [3].

Host defense mechanisms: The ability of *Pseudomonas* species to produce and release metabolites with antagonistic effects on plant pathogens

PGPBs employ various biocontrol mechanisms as part of their antagonistic activities against pathogens. These include the production of antibiotics and lytic enzymes. Antibiotics inhibit or suppress pathogenic bacteria and fungi and are divided into volatile (hydrogen cyanide, 2,3-butanediol, and 6-pentyl- α -pyrone) and non-volatile compounds (phenazines, pyoluteorin, and pyrrolnitrin). The lytic enzymes produced by PGPBs cause the disruption of the cell wall of fungi and oomycetes by degrading the polysaccharides that compose it, such as cellulose (cellulases), chitin (chitinases) and glucan (β -1,3-glucanases) [89-91].

Successful biocontrol, however, hinges on the colonization capacity of the bacteria. Plant roots release various exudates which attract bacteria to the plant's rhizosphere by chemotaxis. The quantity and type of these exudates are influenced by the plant's genetic and environmental factors. Another method of host protection involves the competition for nutrients against pathogenic microorganisms [92].

Several species of *Pseudomonas* promote plant host defense by producing diverse antibacterial and antifungal compounds. Examples include hydrogen cyanide (HCN), pyrrolnitrin, and pyoleutirin produced by *P. aeruginosa* and *P. fluorescens* [93]. Genomic analysis of *P. aeruginosa* FG106 revealed genes related to biocontrol (phenazine biosynthesis, pyrrolnitrin, 2,4-diacetylphloroglucinol (DAPG), HCN). This strain, isolated from tomatoes, showed antagonistic activity against fungal and bacterial pathogens (*Alternaria alternata*, *Botrytis cinerea*, *Clavibacter michiganensis*, *Phytophthora colocasiae*, *Phytophthora infestans*, *Rhizoctonia solani*, and *Xanthomonas euvesicatoria*) by releasing volatile and non-volatile compounds *in vitro*. An inhibitory effect was also observed against *P. infestans* on potato leaves, *B. cinerea* on strawberry leaves, *P. colocasiae* on taro leaves, and against *R. solani* on tomatoes grown in a greenhouse [94]. Evidence that the metabolites of a pathogenic *Pseudomonas* can be an advantageous alternative for tomato production, with productivity gains and reduced use of chemical pesticides.

Bacteria isolated from the maize rhizosphere, characterized as *P. chlororaphis*, were effective against the fungi *A. alternata*, *Colletotrichum dematium*, *C. gloeosporioides*, *Fusarium graminearum*, *F. oxysporum*, *F. subglutinans*, *R. solani*, *Stemphylium lycopersici*, and the oomycete *Phytophthora capsici* *in vitro*. In a greenhouse experiment, *P. chlororaphis* was antagonistic to *R. solani*, which causes the burning of maize leaves and sheaths, when introduced into the seeds and incubated for 12h before planting. The volatile compound 1-undecene and the antibiotic phenazine were identified as biocontrol agents in the bacterial strains analyzed [95].

Regarding the use of lytic enzymes, *P. fluorescens* showed chitinase and β -1, 3-glucanase activities to protect the cumin plant (*Cuminum cyminum* L.) against *F. oxysporum*, which causes wilting in this plant species [96]. *Pseudomonas aeruginosa* produces chitinase, lipase, and protease against root rot pathogens of ginseng (*Panax ginseng*) [97]. *Pseudomonas putida* controls bean rust caused by the fungus *Uromyces appendiculatus* by promoting fungal cell wall rupture under the action of chitinase, lipase, and protease enzymes; among these enzymes, chitinase has a greater contribution to cell disruption [98].

Induced Systemic Resistance (ISR) is an intrinsic plant defense mechanism that is stimulated during interactions between rhizosphere-dwelling PGPBs and their host plants. Unlike Systemic Acquired Resistance (SAR), which involves direct pathogen-plant antagonism often resulting in plant necrosis, ISR is a more sustainable and stable mechanism. It does not harm the plant and, instead, enhances its resistance against a variety of pathogens including bacteria, viruses, fungi, nematodes, and insects [3,89,91].

The ISR activation is primarily mediated through the signaling of the plant hormones jasmonic acid and ethylene, which stimulate defense genes and modify plant metabolism and morphology. This in turn strengthens cell walls, promotes enzyme accumulation and expression of pathogenesis related proteins (PR), and stimulates synthesis of phenolic compounds and phytoalexins. ISR is activated by various bacterial molecules including surface components, volatile and non-volatile metabolites, and secreted compounds such as lipopolysaccharides (LPs), pyocyanins, and siderophores [89,92,99].

Arabidopsis thaliana benefited from the inoculation of *P. fluorescens* PTA-CT2 in its roots when in contact with the necrotrophic fungus *B. cinerea* and the hemibiotrophic bacterium *P. syringae* DC3000. This beneficial effect was observed by ISR, which can be explained by the ability of *P. fluorescens* to induce jasmonic acid/ethylene and NPR1 (non-expressor of PR genes1) signaling pathways. When *Bacillus subtilis* PTA-271 was challenged against the same pathogens, it caused a stronger inhibition against *B. cinerea* than against *P. syringae*, using the same signaling pathway as *P. fluorescens*, but with the addition of salicylic acid, demonstrating that ISR activation pathway depends on the beneficial strain and way of life of the pathogen [100], and that there is a need for further studies to identify the ISR activation pathway by the microorganism. Based on this, targeted tests can be carried out to induce the production of volatile and non-volatile compounds by different species of *Pseudomonas*.

By inducing systemic resistance, *Pseudomonas* spp. contribute to protection against the fungus *Magnaporthe oryzae* in rice [101] and against the bacterium *Clavibacter michiganensis*, which causes canker in tomatoes [102]. When *P. aeruginosa* MF30 is applied to maize (*Zea mays* L.) infected with *R. solani*, Singh and coauthors [103] identified an increase in the expression of PR1 and PR10 genes and a decrease in lesions in leaves and plant sheath, indicating that ISR activation decreased the severity of the disease.

The studies cited serve as examples of the potential for further research using *Pseudomonas* as a protective agent against pathogens in various crops. There are several products registered in Brazil that use only *P. fluorescens* or in combination with *P. chlororaphis*. However, most of them are aimed at insecticides, with few products with fungicide action to combat black scab and damping off. These only use *P. fluorescens* as a microbiological agent [77]. Therefore, advances in research on the antagonistic action of *Pseudomonas* present a significant opportunity for the discovery of new agricultural bioinputs, which may utilize secondary metabolites such as lipopeptides, proven efficient in combating a major disease affecting rice crops (Table

3). The mechanisms described above are summarized in Figure 1. The studies and examples of the described mechanisms are listed in Table 2 and Table 3.

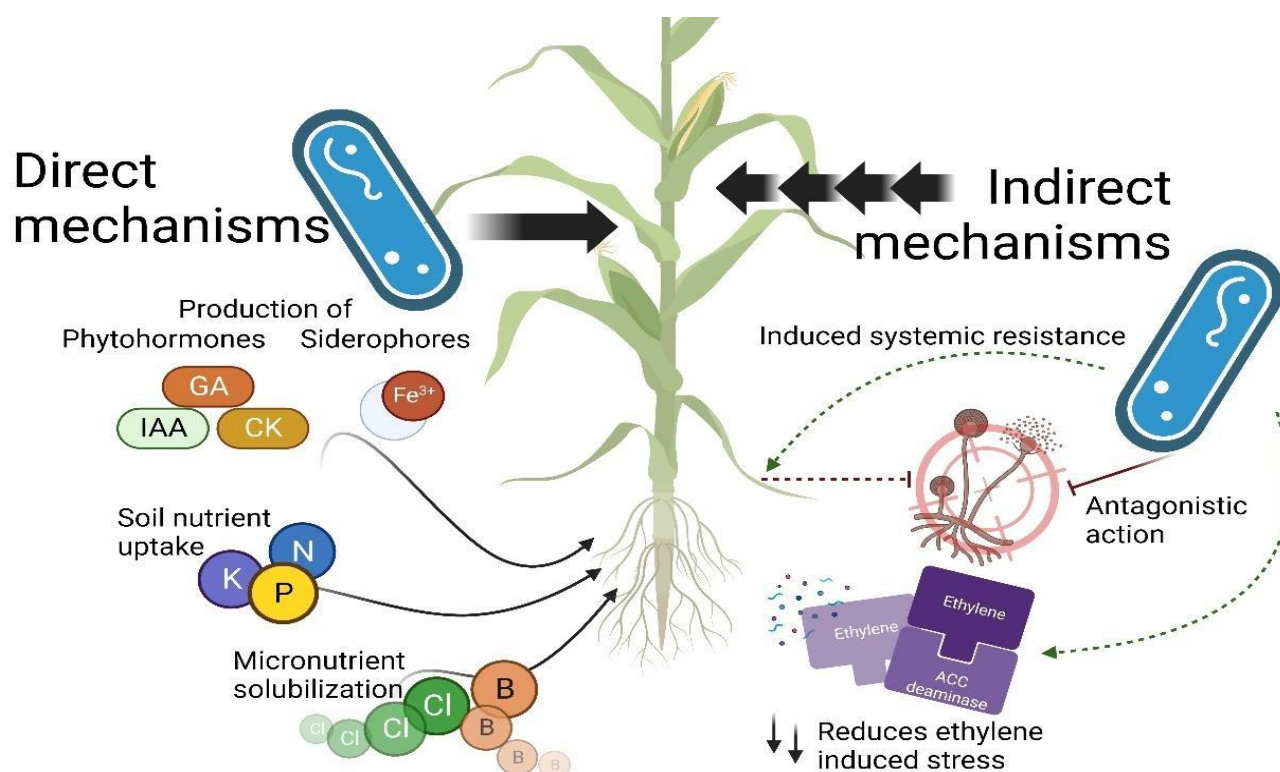


Figure 1. Illustration of Direct and Indirect Mechanisms Leveraged by Endophytic *Pseudomonas* for Plant Growth Promotion. This visual representation outlines the role of endophytic *Pseudomonas* in symbiosis with the host plant, contributing to its growth through a range of mechanisms. Key plant hormones and nutrients implicated in these processes are identified: IAA refers to Indole Acetic Acid, GA denotes Gibberellins, and CK indicates Cytokinins. P, N, and K represent the macronutrients Phosphorus, Nitrogen, and Potassium, respectively, while Cl and B are examples of micronutrients, standing for Chlorine and Boron.

Table 2. Contribution of *Pseudomonas* species in the promotion of plant growth, protection against pathogens and bioremediation by direct mechanisms.

DIRECT MECHANISMS				
Species	Assay	Action Mechanism	Results	References
<i>P. aeruginosa</i>	Mature cow manure	Phosphate-solubilization	Phosphoric nutrients (94.8%) assimilable by plants	[58]
<i>P. aeruginosa</i>	<i>In vitro</i>	Zn-solubilization	Solubilization of zinc a medium containing glucose as a carbon source and ammonium sulphate as nitrogen source	[82]
<i>P. aeruginosa</i>	<i>In vitro</i>	Phosphate-solubilization, IAA syntesis, and siderophore-producing	Plant growth promoting factors (<i>in vitro</i> assays)	[97]
<i>P. azotomorfans</i>	<i>Arabidopsis thaliana</i>	Phosphate-solubilization	Production of 109.47 µl/mL of soluble P, promoting plant growth	[55]
<i>P. azotomorfans</i>	<i>Arabidopsis thaliana</i>	Production of organics acids	Production of malic acid (0.376 µg/mL), lactic acid (0.041µg/mL), and citric acid (0.086 µg/mL)	[55]
<i>P. composti</i>	<i>Suaeda salsa</i>	Mn oxidation	Production of 169µM of Mn oxidation, promoting plant growth	[88]
<i>P. fluorescens</i>	<i>Zea mays</i> L.	Phosphate-solubilization, Siderophore-producing	Increased root growth	[14]
<i>P. fluorescens</i>	<i>In vitro</i>	IAA syntesis	Demonstrates the best carbon and nitrogen sources for induce greater IAA production from <i>P. fluorescens</i>	[65]
<i>P. fluorescens</i>	<i>Arabidopsis thaliana</i>	IAA syntesis	Increased lateral root formation, root hairs, and plant biomass due to auxin induction	[69]
<i>P. fluorescens</i>	<i>Zea mays</i> L.	Siderophore-producing	Increased growth of plant in soil contaminated with cooper (bioremediation)	[79]
<i>P. fluorescens</i>	<i>In vitro</i>	Zn-solubilization	Solubilization of zinc phosphate in a medium containing glucose as a carbon source	[81]
<i>P. japonica</i>	<i>Medicago sativa</i>	Siderophore-producing	Tolerates lead and cadmium (bioremediation) and promotes the survival of plant	[78]
<i>P. jesenii</i>	<i>Cicer arietinum</i>	Phosphate-solubilization	Solubilizing 398.14 µg mL ⁻¹ of P, promoting the germination of 92% of seeds	[56]
<i>P. monteillii</i>	<i>Thlaspi caerulescens</i>	Zn-solubilization	Greater availability of Zn for the plant, which showed greater accumulation of the nutrient in the roots	[85]
<i>P. moraviensis</i>	<i>Triticum aestivum</i> L.	IAA syntesis	Greater than 40% increase in IAA in saline soil and leaves when tryptophan was added	[67]
<i>P. palleroniana</i>	<i>Arabidopsis thaliana</i>	Production of organics acids	Production of oxalic acid (0.216 µg/mL) and malic acid (0.289 µg/mL)	[55]
<i>P. protegens</i> *	<i>Triticale</i> sp., <i>Cucumis sativus</i>	N fixation	Promoted plant growth by expressing the nitrogenase enzyme	[50]
	<i>Arabidopsis thaliana</i>			[51]

Cont Table 2

<i>P. proteolytica</i>	<i>Arabidopsis thaliana</i>	Phosphate-solubilization	Production of 104,61 µl/mL of soluble P, promoting plant growth	[55]
	<i>Arabidopsis thaliana</i>	Production of organics acids	Production of succinic acid (0.059 µg/mL)	[55]
<i>P. putida</i>	<i>Arabidopsis thaliana</i>	IAA syntesis	Increased lateral root formation, root hairs, and plant biomass due to auxin induction	[69]
<i>P. simiae</i> *	<i>Arabidopsis thaliana</i>	Phosphate-solubilization	Hydrolysis of 60% of the phosphate contained in the phytate added to the bioassay	[55]
<i>Pseudomonas</i> sp.	<i>Arabidopsis thaliana</i>	Phosphate-solubilization	Production of 91,84 µl/mL of soluble P, promoting plant growth	[55]
<i>Pseudomonas</i> sp.	<i>Arabidopsis thaliana</i>	Production of organics acids	Production of oxalic acid (0.439 µg/mL)	[55]
<i>Pseudomonas</i> sp.	<i>Oryza sativa</i> , <i>Cynodon dactylon</i> , <i>Poa annua</i>	Phosphate-solubilization	Increased root and shoot growth and formation of root hairs	[9]
<i>Pseudomonas</i> sp.	<i>In vitro</i>	IAA syntesis, phosphate-solubilization, ACC deaminase activity, hydrocarbon degradation	Potential for <i>in vitro</i> plant growth promotion, and <i>in vitro</i> phytoremediation	[10]
<i>Pseudomonas</i> sp.	<i>Oryza sativa</i>	Zn-solubilization	Increased the number of total panicles and grain weight	[83]
<i>Pseudomonas</i> sp.	<i>Zea mays</i> L.	Siderophore-producing	Increased growth of plant in soil contaminated with cooper (bioremediation)	[79]
<i>Pseudomonas</i> sp.	<i>Cicer arietinum</i> L.	Zn-solubilization	Increased zinc absorption, grain yield, and nodule weight and number	[84]
<i>Pseudomonas</i> sp.	<i>Zea mays</i> L.	Zn-solubilization, Mn-solubilization, K-solubilization, N absorption	Increased dry mass and absorption of macro and micro nutrients	[87]
<i>P. stutzeri</i>	<i>Zea mays</i> L.	N fixation	Promoted plant growth and nitrogen accumulation in the plant	[48,49]
<i>P. stutzeri</i>	<i>Allium cepa</i> L.	IAA syntesis	Increased growth and productivity	[70]

*Engineered species.

Table 3. Contribution of *Pseudomonas* species in the promotion of plant growth, protection against pathogens and bioremediation by indirect mechanisms.

INDIRECT MECHANISMS				
Species	Assay	Action Mechanism	Results	References
<i>P. aeruginosa</i>	N.I	Antagonistic action	Production of phenazine, HCN, pyrrolnitrin, and pyoleutirin against pathogenic fungi	[93]
<i>P. aeruginosa</i>	<i>In vitro</i>	Antagonistic action	Production of chitinase, lipase, HCN, and protease against <i>Ilyonectria</i> sp., <i>Neurospora</i> sp., <i>Eutypella</i> sp., <i>Cladosporium</i> sp., <i>Aschersonia</i> sp., and <i>Fusarium</i> sp	[97]
<i>P. aeruginosa</i>	<i>Zea mays</i> L.	Induced Systemic Resistance	Increased the expression of <i>PR1</i> and <i>PR10</i> genes against <i>R. solani</i>	[103]
<i>P. chlororaphis</i>	<i>Zea mays</i> L.	Antagonistic action	Production of undecan, and phenazine against <i>R. solani</i>	[95]
<i>P. fluorescens</i>	N.I	Antagonistic action	Production of phenazine, HCN, pyrrolnitrin, and pyoleutirin against pathogenic fungi	[93]
<i>P. fluorescens</i>	<i>Zea mays</i> L.	Abiotic estress	Increased water deficit stress tolerance	[14]
<i>P. fluorescens</i>	<i>Cuminum cyminum</i> L.	Antagonistic action	Chitinase and β -1, 3-glucanase activities against <i>F.oxysporum</i>	[96]
<i>P. fluorescens</i>	<i>Arabidopsis thaliana</i>	Induced Systemic Resistance	Protection against <i>B. cinere</i> , and <i>P. syringae</i> by induction of jasmonic acid/ethylene and NPR1	[100]
<i>P. protegens</i> *	<i>Triticale</i> sp., <i>Cucumis sativus</i>	Antagonistic action	Protection against <i>R. solani</i>	[50]
<i>P. putida</i>	<i>In vitro</i>	Antagonistic action	Different volatile compounds of pyrazine showed inhibitory activity against <i>P. capsici</i> , <i>P. myriotylum</i> , <i>R. solani</i> , <i>C. gloeosporioides</i> , <i>A. rolfsii</i> , <i>G. moniliformis</i> , <i>M. oryzae</i> , <i>R. pseudosolanacearum</i> , and <i>R. similis</i>	[13]
<i>P. putida</i>	<i>Phaseolus vulgaris</i> L.	Antagonistic action	Promoted the rupture of the cell wall of <i>U. appendiculatus</i> under the action of chitinase, lipase and protease	[98]
<i>Pseudomonas</i> sp.	<i>Oryza sativa</i> , <i>Cynodon dactylon</i> , <i>Poa annua</i>	Antagonistic action	Protection against <i>F.oxysporum</i> , <i>Curvularia</i> sp., and <i>Alternaria</i> sp. by found to produce HCN, and pyrrolnitrin	[9]
<i>Pseudomonas</i> sp.	<i>Oryza sativa</i>	Antagonistic action and Induced Systemic Resistance	Protection against <i>M. oryzae</i> by found to produce cyclic lipopeptides (CLPs)	[101]
<i>Pseudomonas</i> sp.	<i>Solanum lycopersicum</i>	Induced Systemic Resistance	Protection against <i>Clavibacter michiganensis</i> by expressing the <i>PR1a</i> (salicylic acid), and <i>ACO</i> (ethylene) genes	[102]

*Engineered species, N.I.: not informed.

Metabolites, Biotechnological Tools and Application of Bioinputs

The diverse biochemical capabilities of the *Pseudomonas* genus, coupled with their diverse array of metabolite production, have positioned them as promising players in various industries, including agriculture [7]. *Pseudomonas* are attractive microorganisms from an industrial perspective because they grow quickly, utilize readily available carbon sources, and produce enzymes and metabolites applicable to various agricultural applications [104]. Studies have revealed metabolites such as exopolysaccharides generated by *Pseudomonas* species enhancing the yield of sunflower crops in high-salinity terrains [105]. Similarly, volatile metabolites such as 4-nitroguaiacol and quinolone, produced by *P. simiae*, exhibited a positive impact on the growth of soybean plants under saline stress conditions [106]. Short-chain biosurfactants such as rhamnolipids, a metabolite of *P. putida*, demonstrated biotechnological potential, particularly as biocontrol agents and in soil remediation [7]. Although *P. putida* is a pathogenic species, its virulence factors can be modified by metabolic engineering technique those as that performed in *P. aeruginosa* for production of large amounts of phenazines [104,107].

Chromatography and sequential precipitation were used to purify a new laccase enzyme, S2LAC, from marine *Pseudomonas* sp. S2. Laccases are bacterial glycoproteins that can tolerate a wide range of pH values and temperatures, showing great potential for industrial use. In addition, this enzyme demonstrated high efficiency when agricultural waste (potato peel, tea, and wheat flour) was used as substrate. The time required by laccase S2LAC to oxidize organophosphate pesticides was lower than that required by other laccases from different microorganisms [108]. The actions of other *Pseudomonas* metabolites have been extensively discussed by Shahid and coauthors [109], Which highlights the progress in the discovery of new compounds, such as cyclic lipopeptides, aromatic acids, and quorum-sensing signals, in addition to the advancement of genomic databases that include complex metabolic pathways and software, allowing for progress in the study of new metabolites.

The use of biotechnological tools such as genome sequencing may help predict the ability of strains to produce metabolites. However, the expression and production of these metabolites depend on many external factors such as biotic and abiotic stresses. The influence of these factors on bacterial metabolism can also be studied using metabolomic investigations. Rieusset et.al. [110], when investigating the genome of *Pseudomonas* strains, identified 20 gene clusters encoding secondary metabolites involved in plant growth stimulation through siderophore production and biocontrol with the help of antimicrobial compounds such as lankacidin and 2,4-diacetylphloroglucinol [110]. In addition to genome analysis, other omics tools, such as proteomics, metabolomics, fluxomics, and transcriptomics, help in the process of understanding the interactions between cellular components, allowing the production of bacterial compounds to be used more efficiently [7].

The general use of bioinputs will only be attained once their formulation is stable, innovative, and easy to apply on the farm. Several studies have been conducted to develop formulations that maintain viable cells and other biological structures [111]. However, the viability of microorganisms in the field can be reduced due to storage time. In this sense, the use of metabolite-based and cell-free products is a way to address this problem [112], making them an option for using pathogenic *Pseudomonas* in agricultural products.

The method used to apply bioinputs in the field must align the position of the bioinput with its purpose. The product can be diluted and applied during sowing, in the planting furrow, or incorporated into the seeds before sowing, or even solubilized in water for foliar application [113,114].

A variety of bioproducts in the global market use *Pseudomonas* spp. as active compounds. These bioinputs have been validated by the regulatory agencies of the respective countries that have employed them as plant growth promoters and biocontrol agents. Biocontrol products using *P. fluorescens* have been approved in the United States, India, the Netherlands, Australia, and Russia. Other popular registered bioproducts include *P. aurantiaca*, *P. aureofaciens* (USA and Russia), *P. chlororaphis* (Sweden and USA), *P. syringae* (USA), and *Pseudomonas* sp. (Germany) [109,115]. In Brazil, registered products use only *P. fluorescens* with or without *Azospirillum brasilense*. In Brazil, registered products use *P. fluorescens* alone or in consortium with *Azospirillum brasilense*, *Priestia megaterium* and *Bacillus amyloliquefaciens*, and are recommended as inoculants for corn, soybeans, Congo grass (*Brachiaria ruziziensis*) and Asian rice. In products with bioprotection action, there are records containing *P. fluorescens* alone or in consortium with *P. chlororaphis* acting as fungicides and insecticides and *P. oryzihabitans* acting as nematocide [77].

Brazilian Law No. 15,070 of December 2024 establishes the new regulatory framework for the production, use, and commercialization of bioinputs in the country. It establishes that microbial bioinputs are products originating from microorganisms or biotechnological processes and must be registered with the Federal Agricultural Defense Agency, which has the role of inspecting and attesting to the viability and safety of these products [116]. Figure 2 summarizes the pathogenic *Pseudomonas* species and those with

agricultural potential used in this work, as well as those that transit between the two activities. This data demonstrates that the number of species that require further exploration in agriculture necessitates additional applied studies.

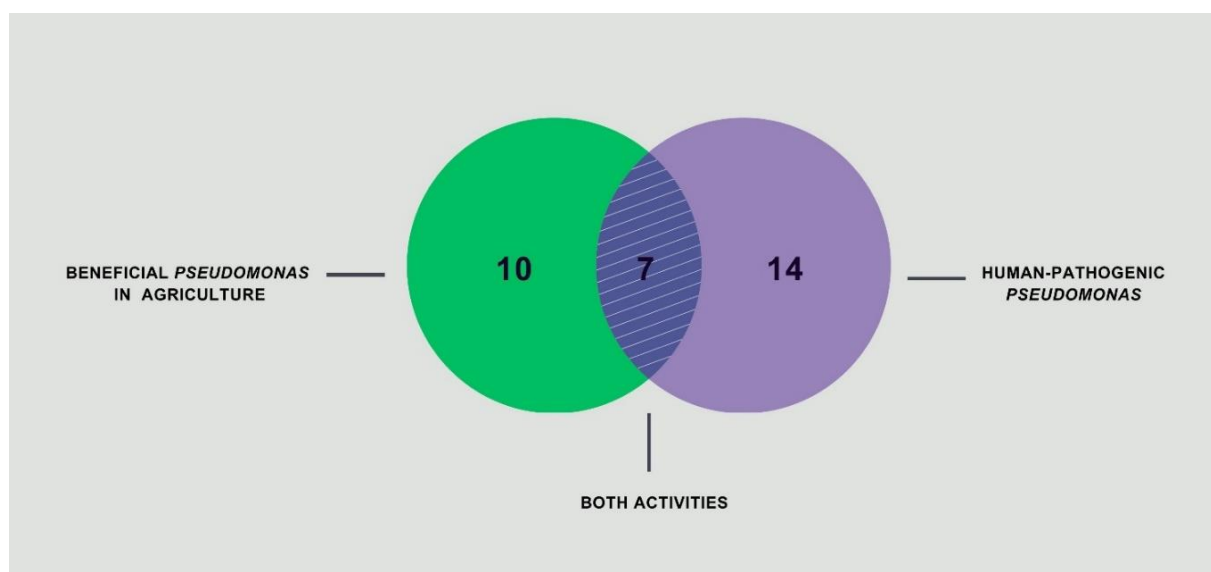


Figure 2. Relationship between species of *Pseudomonas* pathogenic to humans and species beneficial to plants.

Despite the dominance of *P. fluorescens* and *P. syringae* strains, which are deemed low-risk to both human health and the environment [15] commercial products, it's worth noting that even other pathogenic strains of the genus may be leveraged as bioinputs in certain agricultural contexts. These strains can be further manipulated through genetic modifications or by extracting and utilizing their metabolites. Such innovative approaches not only optimize the use of these strains but also ensure safety, posing no harm to consumers or the environment.

The development of new bioproducts involving biotechnological innovations help to gain crop productivity and maintain agricultural sustainability, being important allies against the worrying scenario of desertification of arable areas and the effects of climate change, which can modify the usual planting method of foods. Thus, the search for biological solutions applied to agriculture involving microorganisms is necessary and promising [117].

CONCLUSION

Pseudomonas, a genus boasting hundreds of bacterial species, exhibits a remarkable ability to colonize diverse environments, including the rhizospheric soil and plant roots. The symbiotic interaction between these bacteria and plants can significantly enhance the growth and protection of agricultural crops, thus leading to an improvement in productivity and food quality. Moreover, these bacteria provide an eco-friendlier alternative to chemical products that are traditionally used to address similar agricultural issues. However, given the pathogenic potential of certain species within this genus, both to plants and humans, their utilization is typically restricted by regulatory bodies. To capitalize on the benefits offered by *Pseudomonas* while ensuring safety, the deployment of biotechnological tools proves invaluable. These tools can help identify safe strains, modify or neutralize virulence factors, and even purify and multiply desired metabolites. Consequently, such approaches pave the way for leveraging *Pseudomonas* as plant growth promoters and biocontrol agents, opening up novel and safer avenues for agricultural enhancement.

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